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# Shor's algorithm requires Fanout

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**Abstract**

Shor's algorithm is a canonical quantum supremacy target whose core operation relies on the Quantum Fourier Transform (QFT). In this note, we answer the open question of [FFG+06] by proving that approximating QFT in constant depth, for any $n$-qubit modulus, necessarily requires the $n$-qubit FANOUT operation.

Formally, let $\texttt{QFT}_q$ be the gate acting on $n = \lceil \log q \rceil$ qubits that computes the QFT under modulus $q$. It is known that any $n$-qubit $\texttt{QFT}_q$ can be implemented in constant depth using $\texttt{FANOUT}_n$, i.e. $\texttt{QFT}_q \in \mathsf{QAC}^0_f$ [HS05]. We prove the converse by using a $\texttt{QFT}_q$ gate to construct a state of "non-negligible felinity" [GGJ26a]. Consequently, $\texttt{QFT}_q \in \mathsf{QAC}^0 \iff \texttt{FANOUT}_n \in \mathsf{QAC}^0$.

In the case of $q = 2^n$, such as in Shor's, we approximate $\texttt{FANOUT}_n$ using a single $\texttt{QFT}_{2^n}$ gate and $O(1)$ two-qubit local gates, thus tying the feasibility of realizing Shor's algorithm with NISQ circuits to that of FANOUT.

## 1 Introduction

The Quantum Fourier Transform ($\texttt{QFT}_q$) requires $\Theta(\log n)$ depth using only local gates [CW00]. In 2002, Høyer and Špalek showed that $\texttt{QFT}_q$ can be performed in $O(1)$ depth if this gateset is extended to include FANOUT [HS05]. However, it is yet unknown if $\texttt{QFT}_q$ is implementable by potentially weaker nonlocal interactions. In particular, whether $\texttt{QFT}_q \in \mathsf{QAC}^0$ is open. We show this question is equivalent to Moore's question of whether $\texttt{FANOUT}_n \in \mathsf{QAC}^0$ [Moo99]. This affirmatively resolves the open question posed by Fang, Fenner, Green, Homer and Zhang in 2006: "Is fanout really necessary to do the quantum Fourier transform in constant depth?" [FFG+06].

For additional context on $\mathsf{QAC}^0$ and $\mathsf{QAC}^0_f$, we refer the reader to [GHMP02, Ros21, GMW26] and [GGJ26a].

**Theorem 1.1** (QFT implies Fanout)**.** *For any $q \in \mathbb{N}$ and $n = \lceil \log_2 q \rceil$, if there exists a $\mathsf{QAC}^0$ circuit $C$ approximating $\texttt{QFT}_q$ to a non-negligible fidelity $\delta$, then $\texttt{FANOUT}_n \in \mathsf{QAC}^0$.*

It follows from known results that $\texttt{FANOUT}_n \in \mathsf{QAC}^0 \Rightarrow \texttt{QFT}_q \in \mathsf{QAC}^0$. Although Høyer–Špalek was approximate as stated, it can be made exact by combining with [MZ03] (Appendix A). Thus, as corollary, $\texttt{QFT}_q \in \mathsf{QAC}^0 \iff \texttt{FANOUT}_n \in \mathsf{QAC}^0$.

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**Proof Overview.** The key is to prepare a non-negligible "felinity" state using $C$, which by [GGJ26a] implies exact $\texttt{FANOUT}_n$. The proof starts with a $n+1$-qubit product state whose $\texttt{QFT}_q$ ($2^n \le q < 2^{n+1}$) has constant amplitude on two adjacent frequencies with complementary binary representations. For instance, the frequencies $2^n - 1$ and $2^n$ have binary representations $01\ldots1$ and $10\ldots0$ respectively. Thus, the $\texttt{QFT}_q$ of such a state has constant felinity. To use $C$ in place of $\texttt{QFT}_q$, we first mark the aforementioned branches to isolate them, allowing a $\Omega(\delta^2)$ felinity bound.

Shor's algorithm [Sho97] specifically uses a power-of-two modulus, i.e., a $\texttt{QFT}_{2^n}^\dagger$ [1] gate. Although the above reduction applies, it is technically under $\mathsf{QAC}^0$, not $\mathsf{QNC}^0$. In Lemma 2.3, we describe a circuit for approximating $\texttt{FANOUT}_n$ to arbitrary (depth-dependent) precision using only a $\texttt{QFT}_{2^n}^\dagger$, two-qubit controlled-phases and no extra ancillae. Our circuit effectively approximates a controlled version of Høyer–Špalek's decrement gate [HS05]. We truncate the resulting controlled-single-qubit-unitaries to the most significant qubits to obtain $O(1)$ depth. This lemma ties the realization of Shor's algorithm in the NISQ era (e.g. [ST21]) to that of $\texttt{FANOUT}$.

## 2 The reductions

Following [HS05], define $\omega_q := e^{2\pi i/q}$ and for $x \in \{0, \ldots q-1\}$,

$$\texttt{QFT}_q \left|x\right\rangle := \frac{1}{\sqrt{q}} \sum_{y=0}^{q-1} \omega_q^{xy} \left|y\right\rangle . \tag{2.1}$$

Denote $\widetilde{\texttt{QFT}}_{q,\delta}$ as any gate approximating $\texttt{QFT}_q$ to $\delta$ fidelity, i.e, for all normalized $|\psi\rangle \in \text{span}\{|0\rangle, |1\rangle \ldots |q-1\rangle\}$, $\left|\langle\psi|\texttt{QFT}_q^\dagger \cdot \widetilde{\texttt{QFT}}_{q,\delta}|\psi\rangle\right|^2 \ge \delta$. Note this notion captures a much weaker approximation than the typical $(1-\varepsilon)$ fidelity.

To represent such a *qudit* $|x\rangle$ with $m \ge \lceil \log_2 q \rceil$ *qubits*, we use the binary representation for $x$ with the leftmost qubit being the MSB. For instance, $|2^{m-1}\rangle = |1\rangle |0\rangle^{\otimes m-1}$.

We define $\texttt{FANOUT}_n$ as a unitary map satisfying $|0\rangle_t |0\rangle_A^{\otimes n} \mapsto |0\rangle |0\rangle_A^{\otimes n}$ and $|1\rangle_t |0\rangle_A^{\otimes n} \mapsto |1\rangle |1\rangle_A^{\otimes n}$. Some works refer to a generalized map that applies a $\texttt{CNOT}$ from $t$ to each qubit in $A$. These are closely related, as for any $U$ approximating $\texttt{FANOUT}_n$, a corresponding approximation for generalized fanout exists using $U, U^\dagger$ and local gates [Ros21]. $\texttt{FANOUT}_n \in \mathsf{QAC}^0$ also implies $\texttt{FANOUT}_{\mathsf{poly}(n)} \in \mathsf{QAC}^0$.

We will utilize the following Dirichlet kernel identity. For $D_q(L) := \sum_{r=0}^{L-1} \omega_q^r$,

$$D_q(L) = e^{i\pi(L-1)/q} \frac{\sin \frac{L\pi}{q}}{\sin \frac{\pi}{q}}. \tag{2.2}$$

Finally, we will apply the result of [GGJ26a]: any $\mathsf{QAC}^0$ circuit preparing a $n$-qubit state $\rho$ of felinity $\ge n^{-c}$ for a constant $c$, implies $\texttt{FANOUT}_n \in \mathsf{QAC}^0$. The felinity, $\mathcal{F}_{\boxtimes_n}(\rho)$ is defined as,

$$\mathcal{F}_{\boxtimes_n}(\rho) := 2 \sum_{y \in \{0,1\}^n} \langle y|\rho|y\rangle \cdot \langle y|X^{\otimes n}\rho X^{\otimes n}|y\rangle . \tag{2.3}$$

[1] or $\texttt{QFT}_{2^n}$ depending on convention. The convention doesn't materially affect our proof.

First we show the exact case.

**Lemma 2.1** ($\mathtt{QFT}_q$ produces high felinity)**.** *Let $N := 2^n$, $n \geq 1$, $N \leq q < 2N$ and $y_0, y_1 \in \{0, \dots q-1\}$ such that $y_0 \equiv N \mod q$ and $y_1 \equiv N-1 \mod q$. Then, there is a product state $|\psi_q\rangle$ satisfying, $\left|\langle y_0|\mathtt{QFT}_q|\psi_q\rangle\right| \geq 1/2$ and $\left|\langle y_1|\mathtt{QFT}_q|\psi_q\rangle\right| \geq 1/\pi$.*

*Proof.* The claimed state is,

$$|\psi_q\rangle := |00\rangle \bigotimes_{j=2}^{n} \frac{|0\rangle + \omega_q^{-N2^{n-j}} |1\rangle}{\sqrt{2}} = \frac{1}{\sqrt{N/2}} \sum_{x=0}^{N/2-1} \omega_q^{-Nx} |x\rangle . \tag{2.4}$$

Note the leading 0s ensure the state is only supported on the qudits $\{0, \dots N/2-1\}$ rather than the entire domain of $\mathtt{QFT}_q$, $\{0, \dots q-1\}$. This gives,

$$\mathtt{QFT}_q |\psi_q\rangle = \sqrt{\frac{2}{qN}} \sum_{y=0}^{q-1} \sum_{x=0}^{N/2-1} \omega_q^{(y-N)x} |y\rangle . \tag{2.5}$$

Immediately, $\left|\langle y_0|\mathtt{QFT}_q|\psi_q\rangle\right| = \sqrt{\frac{N/2}{q}} \geq \frac{1}{2}$ because $q \leq 2N$. Also, since $1/2 \geq N/(2q) \geq 1/4$,

$$\left|\langle y_1|\mathtt{QFT}_q|\psi_q\rangle\right| = \left| \sqrt{\frac{2}{qN}} \sum_{x=0}^{N/2-1} \omega_q^{-x} \right| \tag{2.6}$$

$$= \left| \sqrt{\frac{2}{qN}} \cdot D_q(N/2) \right| \tag{2.7}$$

$$= \sqrt{\frac{2}{qN}} \cdot \left| \frac{\sin \frac{\pi N}{2q}}{\sin \frac{\pi}{q}} \right| \tag{2.8}$$

$$\geq \sqrt{\frac{2}{qN}} \cdot \frac{q}{\sqrt{2}\pi} \geq \frac{1}{\pi}. \tag{2.9}$$

□

We will also need the following lemma.

**Lemma 2.2** (Approx. Fourier states $\in \mathsf{QAC}^0$)**.** *For $q$ sufficiently large and $n$ such that $2^n \leq q < 2^{n+1}$, and $y \in \{0, \dots q-1\}$, define $|\phi_y\rangle := \mathtt{QFT}_q^\dagger |y\rangle$. For any $\gamma \leq 1$ such that $\gamma \geq n^{-c}$ for some constant $c$, there exists $|\phi'_y\rangle$ preparable in $\mathsf{QAC}^0$ satisfying, $\left|\langle \phi'_y|\phi_y\rangle\right|^2 \geq 1-\gamma$.*

*Proof.* The state $|\phi_y\rangle = \frac{1}{\sqrt{q}} \sum_{x=0}^{q-1} \omega_q^{-xy} |x\rangle$ when $q = 2^n$ is simply a product state. In fact, for any integer $a$, any $m$-qubit state of the following form is a product state.

$$|\eta_m(a)\rangle := 2^{-m/2} \sum_{x=0}^{2^m-1} \omega_q^{ax} |x\rangle \tag{2.10}$$

When $2^n < q < 2^{n+1}$, first observe for any $\ell = O(\log n)$-qubit register and integer $2^{\ell-1} \leq R \leq 2^\ell$,

the following state is preparable in $\mathsf{QAC}^0$ ($O(1)$ depth, $\mathsf{poly}(n)$ ancillae).

$$|\zeta_\ell(R,a)\rangle := \frac{1}{\sqrt{R}} \sum_{x=0}^{R-1} \omega_q^{ax} |x\rangle \tag{2.11}$$

This is because the comparator $|x\rangle|0\rangle \mapsto |x\rangle|[x<R]\rangle$ has a size-$O(\ell)$ $\mathsf{AC}^0$ circuit and is thus implementable in $\mathsf{QAC}^0$ [GMW26]. Applying it to $|\eta_\ell(a)\rangle|0\rangle$, one obtains,

$$|\zeta'_\ell(R,a)\rangle := \sqrt{\frac{R}{2^\ell}} \cdot |\zeta_\ell(R,a)\rangle|1\rangle + \sqrt{1-\frac{R}{2^\ell}} |\text{bad}\rangle|0\rangle \tag{2.12}$$

which, since $R/2^\ell \geq 1/2$, can be amplified in $O(1)$ depth to $|\zeta_\ell(R,a)\rangle$ (see Cor. 3.16 [GGJ26b]).

Next, set $\ell := \lceil \log_2 \frac{2}{\gamma} \rceil = O(\log n)$ and let $k := n - \ell + 1$. Let $R$ be the integer formed by the $\ell$ most significant bits of $q$, formally $R := \lfloor q/2^k \rfloor$. Since $q \geq 2^n$, $2^{\ell-1} \leq R < 2^\ell$. Then, the desired state is,

$$|\phi'_y\rangle := |\zeta_\ell(R, -2^k y)\rangle \otimes |\eta_k(-y)\rangle \tag{2.13}$$

$$= \frac{1}{\sqrt{2^k R}} \sum_{z=0}^{R-1} \sum_{r=0}^{2^k-1} \omega_q^{-2^k zy - ry} |2^k z + r\rangle \tag{2.14}$$

$$= \frac{1}{\sqrt{q'}} \sum_{x=0}^{q'-1} \omega_q^{-xy} |x\rangle \tag{2.15}$$

where $q' := R \cdot 2^k$. Indeed, this state is preparable in $\mathsf{QAC}^0$ and $\left|\langle\phi_y|\phi'_y\rangle\right|^2 = \frac{q'}{q} \geq 1 - \frac{2^k}{q} \geq 1 - \gamma$. □

**Proof of Theorem 1.1.** Parameterizing as $2^n \leq q < 2^{n+1}$, suppose that $C$ implements $\widetilde{\mathtt{QFT}}_{q,\delta}$ for $\delta \geq n^{-O(1)}$. We prepare a state $|\nu_q\rangle$ in $\mathsf{QAC}^0$ such that $\widetilde{\mathtt{QFT}}_{q,\delta}|\nu_q\rangle$, which is preparable with $C$, has non-negligible felinity on $n$ qubits. Set $y_0, y_1$ as in Lemma 2.1, thus, $|y_1\rangle = |0\rangle|1\rangle^{\otimes n}$, and $|y_0\rangle = |1\rangle|0\rangle^{\otimes n}$ for $q > 2^n$ and $|0\rangle^{\otimes n+1}$ otherwise. If $C$ is exact, $C(|\psi_q\rangle)$ already has the required felinity, otherwise proceed as below.

For any state $|\varphi\rangle$ preparable in $\mathsf{QAC}^0$, it is known that the reflection about it, $I - 2|\varphi\rangle\langle\varphi|$, is also implementable in $\mathsf{QAC}^0$ (Fact 3.9 [GGJ26b]).

Let $A$ be an $n+1$ qubit register and $w_1, w_2$ be two additional qubits. Setting $\gamma := 0.001\delta$, for each $b \in \{0,1\}$, define the following unitaries using the state from Lemma 2.2 and the product state $|+\rangle_{w_1}|b\rangle_{w_2}$.

$$R_b := I - 2|\phi_{y_b}\rangle\langle\phi_{y_b}|_A \otimes |+b\rangle\langle+b|_{w_1,w_2} \tag{2.16}$$

$$R'_b := I - 2|\phi'_{y_b}\rangle\langle\phi'_{y_b}|_A \otimes |+b\rangle\langle+b|_{w_1,w_2} \tag{2.17}$$

Note the operator norm $\|R_b - R'_b\| = 2\sqrt{1 - \left|\langle\phi'_{y_b}|\phi_{y_b}\rangle\right|^2} \leq 2\sqrt{\gamma} \leq 0.1\sqrt{\delta}$, $R'_b$ is implementable in $\mathsf{QAC}^0$, and $R'_0, R'_1$ commute because they reflect about orthogonal states.

Then, for $|\text{EPR}\rangle_{w_1,w_2} = \frac{|00\rangle+|11\rangle}{\sqrt{2}}$, one can also prepare $|\nu_q\rangle := R'_0 R'_1 |\psi_q\rangle_A |\text{EPR}\rangle_{w_1,w_2}$ in $\mathsf{QAC}^0$.

It now suffices to argue that the state $|\psi'\rangle := (\widetilde{\mathtt{QFT}}_{q,\delta})_A \otimes I_{w_1,w_2} |\nu_q\rangle$ has non-negligible felinity. Below we suppress the $I_{w_1,w_2}$ and $\widetilde{\mathtt{QFT}}_{q,\delta}$ implicitly acts only on $A$.

Let $|z_b\rangle := |y_b\rangle_A |(1-b)\rangle_{w_1} |b\rangle_{w_2}$. In binary representation, $z_0, z_1$ are complementary on $\geq n+1$ bits. Observe $R'_{1-b}$ acts as identity on any $|\psi\rangle_{A,w_1} |b\rangle_{w_2}$ and $|1-b\rangle_{w_1} |b\rangle_{w_2}$ is orthogonal to $|\mathrm{EPR}\rangle$ for both values of $b$. Putting it all together,

$$\left|\langle z_b|\widetilde{\mathtt{QFT}}_{q,\delta}|\nu_q\rangle\right| = \left|\langle z_b| \cdot \widetilde{\mathtt{QFT}}_{q,\delta} \cdot R'_b |\psi_q\rangle_A |\mathrm{EPR}\rangle_{w_1,w_2}\right| \tag{2.18}$$

$$\geq \left|\langle z_b| \widetilde{\mathtt{QFT}}_{q,\delta} \cdot R_b \cdot |\psi_q\rangle |\mathrm{EPR}\rangle\right| - \underbrace{\left\|R_b - R'_b\right\|}_{\leq 0.1\sqrt{\delta}} \tag{2.19}$$

$$\geq \left|2 \langle z_b| (\widetilde{\mathtt{QFT}}_{q,\delta})_A \cdot |\phi_{y_b}\rangle\langle\phi_{y_b}|_A |+b\rangle\langle+b|_{w_1,w_2} \cdot |\psi_q\rangle_A |\mathrm{EPR}\rangle_{w_1,w_2}\right| - 0.1\sqrt{\delta} \tag{2.20}$$

$$= \left|2 \langle y_b|\widetilde{\mathtt{QFT}}_{q,\delta}|\phi_{y_b}\rangle \langle\phi_{y_b}|\psi_q\rangle \langle(1-b)b|+b\rangle \langle+b|\mathrm{EPR}\rangle\right| - 0.1\sqrt{\delta} \tag{2.21}$$

$$\geq \left|\langle y_b|\widetilde{\mathtt{QFT}}_{q,\delta} \cdot \mathtt{QFT}_q^\dagger|y_b\rangle\right| \cdot 1/(\pi\sqrt{2}) - 0.1\sqrt{\delta} \tag{2.22}$$

$$\geq 0.11\sqrt{\delta} \tag{2.23}$$

Dropping $w_1, w_2$ and the leftmost qubit $a_0$, $\mathcal{F}_{\bigotimes_n}(\mathrm{tr}_{a_0 w_1 w_2} |\psi'\rangle\langle\psi'|) \geq 4 |\langle z_0|\psi'\rangle|^2 |\langle z_1|\psi'\rangle|^2 = \Omega(\delta^2)$, which is non-negligible. From [GGJ26a], preparing $|\psi'\rangle$ in $\mathsf{QAC}^0$ implies $\mathtt{FANOUT}_n \in \mathsf{QAC}^0$. This concludes the proof.

## 2.1 Local reduction for the Shor modulus.

For a single-qubit unitary $U$, $\mathsf{c}\text{-}U(a,b)$ denotes the controlled $U$ operation with qubit $a$ as the control and $b$ as the target. Then, we show the following depth $k+2$ unitary approximates fanout.

**Lemma 2.3** (Shor Case)**.** *Let $A$ be a set of $n$ qubits representing a qudit register, labeled $a_1, a_2 \ldots a_n$ left to right and let $t$ be a input register and let $N = 2^n$. Define $P_s := \mathrm{diag}(1, e^{-2\pi i/2^s})$ and*

$$U_f^{(k)} := \mathtt{QFT}_N^\dagger(A) \left(\Pi_{s\in[k]} \mathsf{c}\text{-}P_s(t, a_s)\right) H_A^{\otimes n}$$

*Then, for $1 \leq k \leq n$, $U_f^{(k)} |0\rangle_t |0\rangle_A^{\otimes n} = |0\rangle_t |0\rangle_A^{\otimes n}$ and $U_f^{(k)} |1\rangle_t |0\rangle_A^{\otimes n}$ has at least $1 - 4^{1-k}$ fidelity with $|1\rangle_t |1\rangle_A^{\otimes n}$.*

*Proof.* Observe that the actions of $H^{\otimes n}$ and $\mathtt{QFT}_N$ are identical on $|0\rangle^{\otimes n}$. When the input is $|0\rangle$, the map collapses to $\mathtt{QFT}_N^\dagger H_A^{\otimes n} |0\rangle^{\otimes n} = \mathtt{QFT}_N^\dagger \mathtt{QFT}_N |0\rangle^{\otimes n} = |0\rangle^{\otimes n}$, as intended.

Define $|\eta_k\rangle_A := \bigotimes_{s\in[k]} P_s(a_s) \cdot |+\rangle_A^{\otimes n}$. Letting $M := 2^{n-k}$,

$$|\eta_k\rangle = \left(\bigotimes_{s\in[k]} \frac{|0\rangle + \omega_N^{-2^{n-s}} |1\rangle}{\sqrt{2}}\right)_{a_1 \ldots a_k} \otimes |+\rangle_{a_{k+1}\ldots a_n}^{\otimes n-k} \tag{2.24}$$

$$= 2^{-n/2} \sum_{j=0}^{2^k-1} \sum_{r=0}^{M-1} \omega_N^{-Mj} |Mj + r\rangle_A . \tag{2.25}$$

Observe that $|1\rangle_A^{\otimes n} = |N-1\rangle_A$. Therefore, when the input is $|1\rangle$,

$$\langle 1|_t \langle N-1|_A U_f^{(k)} |1\rangle_t |0\rangle_A = \langle N-1|_A \cdot \left(\langle 1|_t \cdot U_f^{(k)} \cdot |1\rangle_t\right) |0^n\rangle_A \tag{2.26}$$

$$= \langle N-1|\mathtt{QFT}_N^\dagger|\eta_k\rangle \tag{2.27}$$

$$= 2^{-n} \sum_{j=0}^{2^k-1} \sum_{r=0}^{M-1} \omega_N^{Mj+r} \omega_N^{-Mj} \tag{2.28}$$

$$= 2^{-n+k} \sum_{r=0}^{M-1} \omega_N^r \tag{2.29}$$

$$= \frac{1}{M} D_N(M) \tag{2.30}$$

Using Eq. (2.2) and third order taylor approximation of $\sin M\theta$ with $\theta = \pi/N$,

$$\left|\langle 1|_t \langle N-1|_A U_f^{(k)} |1\rangle_t |0\rangle_A\right| \geq \frac{1}{M\theta}\left(M\theta - \frac{M^3\theta^3}{6}\right) \tag{2.31}$$

$$\geq 1 - \frac{M^2\pi^2}{6 \cdot N^2} \tag{2.32}$$

$$\geq 1 - \frac{\pi^2}{6 \cdot 4^k} \tag{2.33}$$

Therefore, the fidelity is $\geq 1 - \frac{\pi^2}{3\cdot 4^k} \geq 1 - 4^{1-k}$ as claimed. □

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# A Exact QFT using Fanout

We provide more details on exact version of $\mathtt{QFT}_q \in \mathsf{QAC}^0_f$, which is known folklore.

First note that $\mathsf{QAC}^0_f = \mathsf{QNC}^0_f = \mathsf{QTC}^0$ [HS05, TT16], and thus the $\mathsf{TC}^0$ arithmetic in Høyer–Špalek can be performed exactly. For power-of-two moduli, the only remaining error is in the phase estimation part. This is exactly what [MZ03] addresses by making this error uniform and removing it with exact amplitude amplification.

For non-power-of-two moduli, the missing piece is an exact preparation of $|\phi_0\rangle := \frac{1}{\sqrt{2}} \sum_{x=0}^{q-1} |x\rangle$. One can prepare $|\phi_0\rangle$ in $\mathsf{QAC}^0_f$ as follows: (1) Mark the $x \leq q$ branches of $|+\rangle^{\otimes \lceil \log_2 q \rceil}$ using $n$-bit comparator (available in $\mathsf{AC0}/\mathsf{TC0}$), (2) perform exact amplitude amplification to get $|\phi_0\rangle$ in $O(1)$ depth. This is same as the first part of Lemma 2.2 but with $n$-bit comparators.